

Activity 1: Take three integers and display, the smallest and largest among them, without using standard library functions

```
import java.util.Scanner;

public class SmallestAndLargest {
    public SmallestAndLargest() {

        public static void main(String[] var0) {
            Scanner var1 = new Scanner(System.in);
            System.out.println("Enter the first number: ");
            int var2 = var1.nextInt();
            System.out.println("Enter the second number: ");
```

```
int var3 = var1.nextInt();
System.out.println("Enter the third number: ");
int var4 = var1.nextInt();
int var5 = var2;
int var6 = var2;
if (var3 < var2) {
    var5 = var3;
}
if (var4 < var5) {
    var5 = var4;
}
if (var3 > var2) {
    var6 = var3;
}
if (var4 > var6) {
    var6 = var4;
}
System.out.println("The smallest number is: " + var5);
System.out.println("The largest number is: " + var6);
var1.close();
}
```

Activity 2:

Take a number and print all the prime numbers up to that number, without using standard library functions

```
import java.util.Scanner;

public class PrimeNumbers {
    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter a number: ");
        int n = scanner.nextInt();

        System.out.println("Prime numbers up to " + n + ":");
        for (int i = 2; i <= n; i++) {
            if (isPrime(i)) {
                System.out.print(i + " ");
            }
        }
        scanner.close();
    }

    public static boolean isPrime(int num) {
        if (num < 2) {
            return false;
        }
        for (int i = 2; i <= num / 2; i++) {
            if (num % i == 0) {
                return false;
            }
        }
        return true;
    }
}
```

Activity 3:

Take a number and display the sum of even numbers and sum of odd numbers up to that number, without using standard library functions

```
import java.util.Scanner;

public class SmallestAndLargest {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter the first number: ");
        int num1 = scanner.nextInt();
        System.out.println("Enter the second number: ");
        int num2 = scanner.nextInt();
        System.out.println("Enter the third number: ");
        int num3 = scanner.nextInt();
        int smallest = num1;
```

```
int largest = num1;

    if (num2 < smallest) {
        smallest = num2;
    }
    if (num3 < smallest) {
        smallest = num3;
    }
    if (num2 > largest) {
        largest = num2;
    }
    if (num3 > largest) {
        largest = num3;
    }
    System.out.println("The smallest number is: " + smallest);
    System.out.println("The largest number is: " + largest);
    scanner.close();
}
```

2.Convert the string input taken using the Scanner class to double datatype using parseDouble

```
import java.util.Scanner;

public class StringToDouble {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter a number as a string: ");

        String input = scanner.nextLine();

        try {

            double number = Double.parseDouble(input);

            System.out.println("The double value is: " + number);

        } catch (NumberFormatException e) {

            System.out.println("Invalid input. Please enter a valid number.");

        }

        scanner.close();

    }

}
```

3. generate a random number between 0 and -100 using the inbuilt random function

```
import java.util.Random;

public class RandomNumberGenerator {

    public static void main(String[] args) {

        Random random = new Random();

        int positiveRandom = random.nextInt(101);

        int negativeRandom = -positiveRandom;

        System.out.println("Random number between 0 and -100: " + negativeRandom);

    }

}
```

- 4: use string class techniques to concatenate three input taken from the keyboard and print each of its length?

```
import java.util.Scanner;

public class StringConcatenation {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter the first string: ");
        String str1 = scanner.nextLine();

        System.out.println("Enter the second string: ");
        String str2 = scanner.nextLine();

        System.out.println("Enter the third string: ");
        String str3 = scanner.nextLine();

        String concatenatedString = str1 + str2 + str3;

        System.out.println("length of first string: " + str1.length());
        System.out.println("length of second string: " + str2.length());
        System.out.println("length of third string: " + str3.length());
        System.out.println("Concatenated string: " + concatenatedString);
        System.out.println("length of concatenated string: " + concatenatedString.length());

        scanner.close();
    }
}
```